

Tianyi Xu

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Education

University of Wisconsin

Set 2021 – Dec 2025

BS in Computer Science, Data Science, Mathematics

- GPA: 3.944/4.000
- **Coursework:** Foundation Models, Deep Learning in Computer Vision, Learning Based Image Synthesis and Manipulation, Data-Centric ML, Real Analysis, Probability Theory, Theoretical Data Science

Research Experiences

Research Assistant

Madison, WI

Advisor: Junjie Hu

July 2025 – Present

- Learn robust Hmong word representations that are **speaker- and tone-invariant** by adapting a multilingual speech encoder with Wav2vec, XLS-R, and Fairseq in a low-resource setting.
- Reduced tone/gender bias with **contrastive learning** on word-level pairs (cross-gender same-word and tone variants as positives), with evaluation via **cross-speaker/tone retrieval** (Top-1/Top-5) and targeted data augmentation.

Research Intern @ SJTU Epic Lab

Remote

Advisor: Linfeng Zhang, Mentor: Shaobo Wang

Aug 2025 – Present

- Built a **multi-GPU** GPT-2 pretraining pipeline (distributed training, eval, reproducible code).
- Integrated **PiKE** for online domain-mix adaptation and extended **GREATS** to work with **AdamW** and **Muon** for data selection.
- Ran ablation studies on data mixes/selection cadences; evaluated on **MMLU**, **HellaSwag**, **OpenBookQA**, etc, improved accuracy for **2% - 3%** on average.

Research Assistant

Madison, WI

Advisor: Pedro Morgado

Jan 2025 – May 2025

- Built a **latent generative pipeline** for vision-guided audio separation (VAE latent space) that cuts compute/memory while preserving quality.
- Implemented and compared **latent diffusion** and **rectified flow matching** with visual conditioning; introduced **per-channel z-norm + clipping** that stabilized training.
- Built a visual-guided audio diffusion pipeline with U-Net conditioning; evaluated with FAD, LPAPS, CLAP, and ImageBind on AVE/Music datasets.

Research Assistant

Madison, WI

Advisor: Claudia Solis-Lemus

Jan 2023 – Present

- Analyzing **large, high-dimensional microbiome datasets** to identify patterns, employing **visualization, dimension reduction, statistical analysis, regression models**, and **network analysis** in **R**.
- Working on a project for **automatic biodiversity detection**. Designed a **label-efficient bioacoustic detector**: self-supervised **AudioMAE** pretraining reused as a detector backbone, achieving **~0.9 AP** with strong cross-site generalization.
- Built supervised baselines and a reproducible pipeline with **Faster R-CNN**, **YOLO**, and **FCOS** in **PyTorch/librosa/h5py** for end-to-end benchmarking.

Publications

Combined effects of methyl bromide and soil amendments on soil bacterial and fungal communities in turfgrass

In Submission

Tianyi Xu*, Salma Mukhtar*, Evan Gorstein, Claudia Solis-Lemus, Ming Yi Chou, Paul Koch

In submission to *Environmental Microbiology*. *Equal contribution.

Stem rot affects the structure of rhizosphere microbiome in berseem clover (*Trifolium alexandrinum*) (Published on Rhizosphere)

Apr 2025

Salma Mukhtar, Zain Ahmad, Noor Khan, *Tianyi Xu*, Dalaq Aiysha

[10.1016/j.rhisph.2025.101073](https://doi.org/10.1016/j.rhisph.2025.101073) [🔗](#)

BiomedBank: A Large-scale, Multimodal Data Ecosystem for Advancing Biomedical AI

In preparation

TBD

Manuscript in preparation.

Talks / Presentations

Undergraduate Research Symposium – Poster Presentation

Apr 2025

27th Annual Undergraduate Symposium, University of Wisconsin–Madison

"Biodiversity Detection Using Acoustic Signals with Deep Learning"

Presented as part of the requirement of Holstrom Environmental Fellowship on the progress of the project.

[Abstract Link \(Page 164\)](#) [🔗](#)

Undergraduate Research Symposium – Poster Presentation

Apr 2024

26th Annual Undergraduate Symposium, University of Wisconsin–Madison

"Biodiversity Detection Using Acoustic Signals with Deep Learning"

[Abstract Link \(Page 139\)](#) [🔗](#)

Awards

Holstrom Environmental Research Fellowship

2024

Awarded \$3,000 + \$1,000 (advisor) for research on acoustic event detection in tropical rainforests using deep learning.

UW–Madison Summer Scholarship

2024

Awarded \$1,000 for Summer 2024 studies in recognition of outstanding academic achievement.

Dean's List (Six Semesters)

2021–2024

Recognized for maintaining a GPA above 3.85 in Fall 2021, Spring 2022, Fall 2022, Fall 2023, Spring 2024, and Fall 2024.

Other Experiences & Projects

Project Assistant / Intern *Remote*

Nov 2024 – May 2025

- Contributed to *BiomedBank*, a large-scale multimodal biomedical dataset project, collaborated with Jiong-xiao Wang(UW-Madison), Kai Zhang(Lehigh), Prof. Lichao Sun(Lehigh), and Prof. Chaowei Xiao(UW-Madison/JHU).
- Conducted literature review on data quality and built preprocessing/deduplication pipeline for biomedical NLP. Benchmarked biomedical QA and Clinical Trial Dataset with VLLM deployed medical/general Large Language Models (e.g Med-LlaMa3, GPT-OSS, Qwen3).

Project Assistant *Wisconsin Alzheimer's Disease Research Center*

Sept 2023 – Aug 2024

- Developed diagnosis prediction pipeline with SMOTE, Random Forest, and XGBoost.
- Explored CNN and phylogenetic embeddings for feature representation; used VAE for data synthesis.

Peer Mentor *UW-Madison Department of Computer Science*

Sept 2022 – May 2024

- Mentored 400+ students in CS 220, 320, and 540 on Python, AI, and data science topics.
- Hosted 400+ office hours covering OOP, visualization, and ML/DL fundamentals.

Developer *UW-Madison DoIT Helpdesk*

Apr 2022 – Aug 2022

- Improved UI/UX with dark mode enhancements and user preference handling.
- Built and documented APIs integrated with MySQL for efficient data access.

- Created staff filtering system in React.js to support searches across 1000+ employees.

Selected Course Projects/Papers *University of Wisconsin-Madison*

2023-2024

- *ABTUNE – Alpha Batch Size Tuning (Graduate Foundation Model Course, Full Mark)*: Benchmarked two-phase training and adaptive batch size strategies on noisy vs. clean data across image and text classification tasks.
- *Learning Smartly – Data-Centric Methods for LMs (50% Graduate-Level Data-Centric ML Course, Full Mark)*: Developed a hybrid framework combining LLM-generated label functions and active learning to improve weak supervision in text classification tasks.
- *Image Style Transfer with Diffusion Models (STAT 453 Deep Learning, A)*: Supervised fine-tuned (SFT) Stable Diffusion using curated paired datasets (CycleGAN, AdaIN, GPT-4 prompts) for semantic-preserving, text-guided multi-style transfer.
- *ImageEditNet – SFT on InstructPix2Pix (Graduate Learning-Based CV Course, A)*: Improved real-image text-guided editing via SFT on MagicBrush and GAN-generated datasets, boosting CLIP similarity and facial editing quality.
- *ASL Fingerspelling Recognition (CS 539 Neural Network, Full Mark)*: Fine-tuned Faster R-CNN (ResNet50, MobileNetV2) for real-time ASL letter detection, achieving 0.913% IoU and 0.7% accuracy; deployed webcam demo for live inference.

Skills

Programming Languages: Python, R, Java, C/C++, JavaScript, HTML, CSS, SQL

Machine Learning / Data Science: PyTorch, TensorFlow, Scikit-Learn, Hugging Face Transformers, Diffusers, PyTorch Geometric, Pandas, Spark, HDFS, BigQuery, Cassandra, Kafka, CUDA, Wandb, Fairseq

Development Tools: Docker, Flask, Selenium, MySQL, SQLite3, Git, Linux, GCP, JUnit, Vim, VS Code, PyCharm, Snowflake, Elasticsearch, Streamlit, Airbyte

Web Technologies: React, Node.js

Natural Languages: Chinese Mandarin (native), English (fluent), Japanese (familiar)